

Supplementary Material

MTGDriver: Multitask Graph Learning for Cancer Driver Gene Prioritization

1 Training and implementation details

The hyperparameter search space comprises the number of GCN layers $\{1, 2\}$, hidden dimensions $\{64, 128\}$, dropout rates $\{0.3, 0.4, 0.5\}$, learning rates $\{10^{-2}, 3 \times 10^{-3}, 10^{-3}\}$, and weight decay values $\{10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$. The configuration yielding the highest validation AUPRC is selected for each training fold.

Dropout is incorporated in both the encoder and task-specific classifiers. GCN layers utilize Xavier initialization¹, while linear layers employ Kaiming initialization².

MTGDriver is implemented in Python 3.11.13³ using PyTorch⁴ and PyTorch Geometric⁵. The graph convolutional layers are implemented using GCNConv, following the GCN formulation in⁶. All experiments are performed on a single GPU.

2 Extended Hits@K evaluation

To provide a more complete evaluation of top-ranked cancer driver gene recovery, we additionally report Hits@K for $K \in \{1, 3, 5, 10\}$ across all 12 TCGA cancer types. These results extend the main-text analysis, which focuses on $K \in \{1, 3, 5\}$, by including a broader ranking cutoff at $K = 10$.

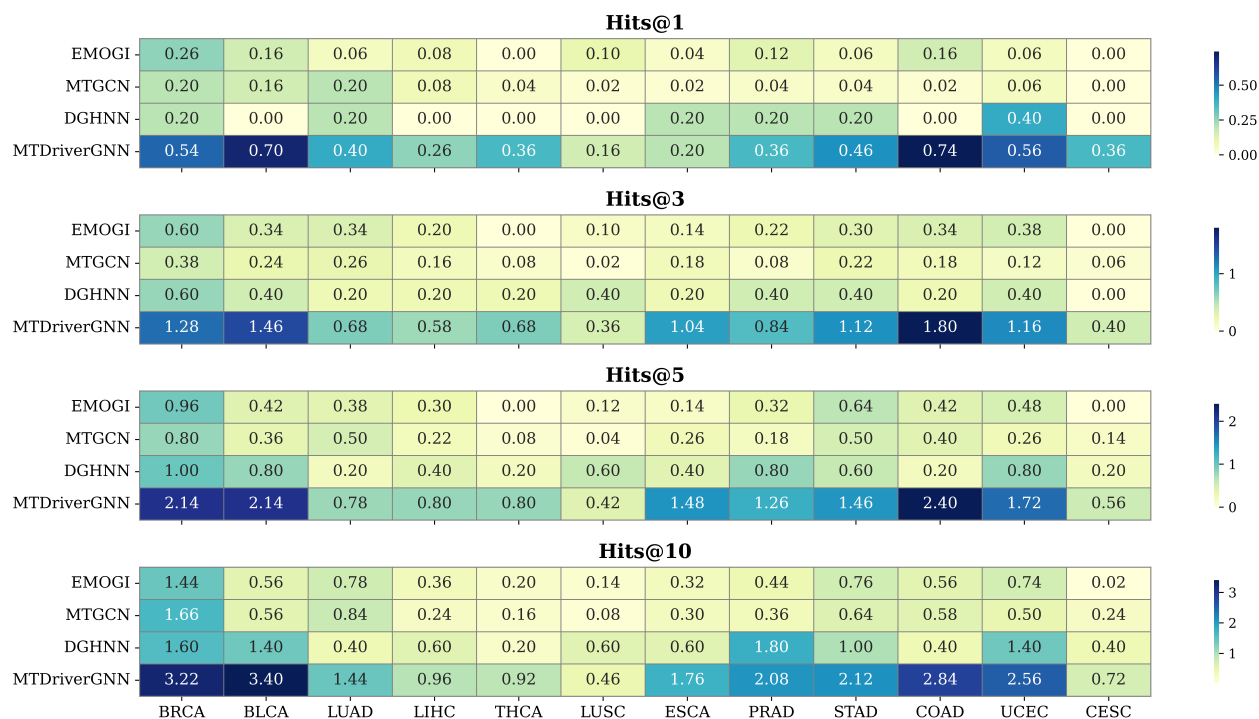


Figure S1. Extended Hits@K evaluation of MTGDriver and baseline methods across 12 TCGA cancer types for $K \in \{1, 3, 5, 10\}$. Each heatmap reports the mean number of known cancer driver genes recovered among the top-K ranked predictions for EMOGI, MTGCN, DGHNN, and MTGDriver. Values represent averages over 10 independent runs with stratified 5-fold cross-validation.

Table S1. Comparison of MTGDriver with baseline methods across 12 cancer types, evaluated using AUPRC (mean $\pm$ standard deviation).

Cancer	MTGDriver	EMOGI	MTGCN	DGHNN
BRCA	0.7306 $\pm$ 0.0096	0.6482 $\pm$ 0.0045	0.6583 $\pm$ 0.0040	0.6832 $\pm$ 0.0062
BLCA	0.7155 $\pm$ 0.0219	0.5485 $\pm$ 0.0101	0.6568 $\pm$ 0.0077	0.6935 $\pm$ 0.0064
LUAD	0.6653 $\pm$ 0.0096	0.5591 $\pm$ 0.0105	0.6279 $\pm$ 0.0099	0.6410 $\pm$ 0.0091
LIHC	0.5076 $\pm$ 0.0196	0.3845 $\pm$ 0.0193	0.4645 $\pm$ 0.0195	0.4811 $\pm$ 0.0191
THCA	0.3533 $\pm$ 0.0153	0.2587 $\pm$ 0.0084	0.2907 $\pm$ 0.0110	0.3145 $\pm$ 0.0233
LUSC	0.4461 $\pm$ 0.0536	0.2222 $\pm$ 0.0266	0.3099 $\pm$ 0.0407	0.4054 $\pm$ 0.0542
ESCA	0.5027 $\pm$ 0.0289	0.4174 $\pm$ 0.0297	0.4721 $\pm$ 0.0279	0.4763 $\pm$ 0.0145
PRAD	0.7468 $\pm$ 0.0256	0.5233 $\pm$ 0.0220	0.6143 $\pm$ 0.0189	0.7058 $\pm$ 0.0129
STAD	0.7240 $\pm$ 0.0168	0.4604 $\pm$ 0.0153	0.5931 $\pm$ 0.0117	0.6449 $\pm$ 0.0083
COAD	0.5353 $\pm$ 0.0205	0.3459 $\pm$ 0.0067	0.3816 $\pm$ 0.0085	0.4468 $\pm$ 0.0220
UCEC	0.7121 $\pm$ 0.0235	0.4020 $\pm$ 0.0143	0.4954 $\pm$ 0.0145	0.5342 $\pm$ 0.0250
CESC	0.5715 $\pm$ 0.0687	0.5547 $\pm$ 0.0522	0.5972 $\pm$ 0.0528	0.5635 $\pm$ 0.0414

Table S2. Performance of Single-task and Multi-task settings across 12 cancer types, evaluated using AUPRC (mean $\pm$ standard deviation).

Cancer	Single-task	Multi-task
BRCA	0.6233 $\pm$ 0.0100	0.7110 $\pm$ 0.0125
BLCA	0.7104 $\pm$ 0.0114	0.7028 $\pm$ 0.0191
LUAD	0.6473 $\pm$ 0.0052	0.6508 $\pm$ 0.0157
LIHC	0.4556 $\pm$ 0.0209	0.4863 $\pm$ 0.0226
THCA	0.2435 $\pm$ 0.0144	0.3107 $\pm$ 0.0309
LUSC	0.3054 $\pm$ 0.0377	0.4197 $\pm$ 0.0402
ESCA	0.4000 $\pm$ 0.0429	0.4926 $\pm$ 0.0186
PRAD	0.6577 $\pm$ 0.0390	0.7021 $\pm$ 0.0315
STAD	0.6796 $\pm$ 0.0261	0.6938 $\pm$ 0.0235
COAD	0.3978 $\pm$ 0.0237	0.5136 $\pm$ 0.0212
UCEC	0.5750 $\pm$ 0.0190	0.6837 $\pm$ 0.0238
CESC	0.5098 $\pm$ 0.0521	0.4628 $\pm$ 0.0281

Table S3. Comparison of MTGDriver without and with pretraining across 12 cancer types, evaluated using AUPRC (mean $\pm$ standard deviation).

Cancer	W/O Pretraining	Pretraining
BRCA	0.7110 $\pm$ 0.0125	0.7306 $\pm$ 0.0096
BLCA	0.7028 $\pm$ 0.0191	0.7155 $\pm$ 0.0219
LUAD	0.6508 $\pm$ 0.0157	0.6653 $\pm$ 0.0096
LIHC	0.4863 $\pm$ 0.0226	0.5076 $\pm$ 0.0196
THCA	0.3107 $\pm$ 0.0309	0.3533 $\pm$ 0.0153
LUSC	0.4197 $\pm$ 0.0402	0.4461 $\pm$ 0.0536
ESCA	0.4926 $\pm$ 0.0186	0.5027 $\pm$ 0.0289
PRAD	0.7021 $\pm$ 0.0315	0.7468 $\pm$ 0.0256
STAD	0.6938 $\pm$ 0.0235	0.7240 $\pm$ 0.0168
COAD	0.5136 $\pm$ 0.0212	0.5353 $\pm$ 0.0205
UCEC	0.6837 $\pm$ 0.0238	0.7121 $\pm$ 0.0235
CESC	0.4628 $\pm$ 0.0281	0.5715 $\pm$ 0.0687

Table S4. Summary statistics of feature counts and gene counts across 12 cancer types.

	BRCA	BLCA	LUAD	THCA	LIHC	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
Mutation-derived features	4	4	4	4	4	4	4	4	4	4	4	4
Pathway-derived features	8	6	7	5	11	7	5	8	8	9	8	14
Total features	12	10	11	9	15	11	9	12	12	13	12	18
Driver genes	202	95	179	64	82	26	71	50	73	141	68	17
Passenger genes	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187

Table S5. Over-Representation Analysis (ORA) for BRCA and PRAD using representative cancer-related HDO and KEGG terms.

Cancer	Collection	Top-K	Count	Fold enrichment	FDR (BH)
BRCA	HDO	100	17	3.38	1.55×10^{-4}
		150	23	3.01	3.88×10^{-5}
		200	27	2.73	3.25×10^{-5}
	KEGG	100	8	6.14	1.30×10^{-4}
		150	11	5.41	1.25×10^{-5}
		200	16	5.99	2.89×10^{-8}
PRAD	HDO	50	13	6.09	2.70×10^{-5}
		100	26	6.28	5.47×10^{-12}
		150	37	6.07	2.17×10^{-12}
		200	47	5.81	1.61×10^{-21}
	KEGG	50	15	28.42	9.52×10^{-17}
		100	30	27.37	4.80×10^{-35}
		150	32	19.71	2.77×10^{-32}
		200	39	17.57	3.01×10^{-38}

3 Additional biological relevance analysis

3.1 Over-representation analysis

Supplementary Table S5 provides the complete ORA results across the evaluated ranking cutoffs. For BRCA, enrichment of the representative HDO breast cancer term remains significant from the top 100 to the top 200 predictions, with fold enrichment ranging from 3.38 to 2.73. The KEGG breast cancer pathway shows stronger enrichment, ranging from 5.99- to 6.14-fold across the same ranking depths. For PRAD, the representative HDO prostate cancer term shows approximately six-fold enrichment across the top 50–200 predictions. The strongest enrichment is observed for the KEGG prostate cancer pathway, reaching 28.42-fold at the top 50 and remaining strongly enriched at the top 200 predictions (17.57-fold). All reported associations remain significant after Benjamini–Hochberg correction for multiple testing.

3.2 Independent disease-gene validation

Supplementary Table S6 provides the detailed validation results against DisGeNET and Open Targets. In DisGeNET, PRAD shows 6.01-fold enrichment at both the top 50 and top 100 predictions, with Precision@K of 0.22 at both cutoffs. BRCA shows 3.92-fold enrichment at the top 50 and 3.48-fold enrichment at the top 100. Evaluation across the complete rankings yields AUROC/AUPRC values of 0.645/0.081 for PRAD and 0.612/0.074 for BRCA.

Open Targets provides stronger independent support. At an association-score threshold of 0.20, BRCA achieves enrichment values of 3.29, 3.11, and 2.67 at the top 50, 100, and 200 predictions, respectively, with an overall AUROC of 0.707 and AUPRC of 0.417. At the more stringent threshold of 0.40, enrichment increases to 5.70 at both the top 50 and top 100 predictions.

For PRAD, the Open Targets threshold of 0.20 yields particularly strong enrichment, reaching 7.46-fold at the top 50, 8.34-fold at the top 100, and 7.46-fold at the top 200 predictions. The corresponding complete-ranking AUROC and AUPRC are 0.785 and 0.186, respectively. At the more stringent threshold of 0.40, relatively few supported genes are available among the highest-ranked predictions, although the complete ranking achieves an AUROC of 0.862.

Table S6. Independent disease-gene validation for BRCA and PRAD using DisGeNET and Open Targets. DisGeNET results use a gene-disease association score threshold of 0.40, while Open Targets results use global association-score thresholds of 0.20 and 0.40.

Cancer	Score	K	Count	P@K	Enrichment	FDR (BH)	AUROC	AUPRC
DisGeNET								
BRCA	0.40	50	9	0.180	3.92	4.80×10^{-4}	–	–
		100	16	0.160	3.48	1.83×10^{-5}	–	–
		200	24	0.120	2.61	2.08×10^{-5}	–	–
		All	–	–	–	–	0.612	0.074
PRAD	0.40	50	11	0.220	6.01	1.75×10^{-6}	–	–
		100	22	0.220	6.01	1.05×10^{-11}	–	–
		200	33	0.165	4.51	3.99×10^{-13}	–	–
		All	–	–	–	–	0.645	0.081
Open Targets								
BRCA	0.20	50	37	0.74	3.29	1.60×10^{-14}	–	–
		100	70	0.70	3.11	5.58×10^{-24}	–	–
		200	120	0.60	2.67	1.91×10^{-30}	–	–
		All	–	–	–	–	0.707	0.417
	0.40	50	12	0.24	5.70	9.26×10^{-7}	–	–
		100	24	0.24	5.70	3.28×10^{-12}	–	–
		200	34	0.17	4.04	3.28×10^{-12}	–	–
		All	–	–	–	–	0.725	0.105
PRAD	0.20	50	17	0.34	7.46	3.61×10^{-11}	–	–
		100	38	0.38	8.34	1.61×10^{-25}	–	–
		200	68	0.34	7.46	1.67×10^{-41}	–	–
		All	–	–	–	–	0.785	0.186
	0.40	50	1	0.02	8.76	1.19×10^{-1}	–	–
		100	1	0.01	4.38	2.05×10^{-1}	–	–
		200	5	0.03	10.95	1.22×10^{-4}	–	–
		All	–	–	–	–	0.862	0.020

For all ranking-depth enrichment analyses in Supplementary Table S6, statistical significance is assessed using a one-sided hypergeometric test followed by Benjamini–Hochberg correction for multiple testing.

References

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